

# Trace Boggs

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## EDUCATION

### The University of Oklahoma

*M.S. Computer Science*

Expect. Grad 2028

*Norman, OK*

### Cameron University

*B.S. Computer Science & B.A. Mathematics, History Minor | GPA: 4.0/4.0*

May 2026

*Lawton, OK*

## WORK EXPERIENCE

### Software Engineering Intern

*Brunswick - Navico Group*

May 2025 – Aug. 2025

*Tulsa, OK*

- Developed a **Python** desktop GUI app for automated integration and system testing of embedded products via **BLE**, including device configuration, peripheral connectivity, firmware responses, and functional validation.
- Implemented **real-time test monitoring** and automated logging.
- Reduced engineer regression-testing time by **95%** during development sprints.

### Mathematics Tutor

*Cameron University*

Jan. 2024 – May. 2026

*Lawton, OK*

- Tutored **College Algebra, Trigonometry, Calculus, Discrete Mathematics, Data Structures, and Algorithms** through individualized and group sessions, adapting explanations and problem-solving strategies to students' learning gaps.

## RESEARCH EXPERIENCE

### Machine Learning Supervised Research

*CU Dept. Computing and Mathematical Sciences*

Jan. 2025 - Dec. 2025

*Lawton, OK*

- Investigated **real-time facial emotion recognition** using **CNNs, transfer learning, and MobileNetV2** trained on the FER-2013 dataset.
- Evaluated **static-image vs. real-time inference** with OpenCV and explored **model calibration, temporal smoothing, and lightweight inference** to improve deployment robustness.

### Math Capstone Iterative Methods Research

*CU Dept. Computing and Mathematical Sciences*

Aug. 2025 - May 2026

*Lawton, OK*

- Conducted **numerical analysis** research through proof development, example construction, technical writing, and figure design.
- Co-authored research on **local convergence of Newton-like methods in normed linear spaces**; manuscript submitted to *Mathematical Methods in the Applied Sciences*.
- Presented research at **NCUR 2026**, the National Conference for Undergraduate Research.

## PROJECTS - (FOR FULL PORTFOLIO: [HTTPS://TBOGGS05.GITHUB.IO/](https://tboggs05.github.io/))

### LightStop Mobile Game

Feb. 2023

- Developed and released a mobile game for **Android and iOS** using **Unity and C#**, designed for replayability and compatibility with older devices.
- Implemented **live leaderboards, Unity Ads, reusable gameplay components, game-state management, and input handling**.

### Euler's Method ODE Calculator Webapp

Sept. 2024

- Built a full-stack web application for numerically solving **ordinary differential equations** using **Euler's Method**, supporting user-defined ODEs and initial conditions.
- Implemented a **C++ REST backend**, custom input parsing, and client-server communication; deployed the application on a **Raspberry Pi**.
- Configured **Nginx reverse proxying and SSL** for secure public access.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, C#, Java, JavaScript, SQL, HTML/CSS, LaTeX

**Libraries/Frameworks:** TensorFlow, Pandas, Matplotlib, OpenCV, Unity, Flutter, Node.js

**Systems/Tools:** Linux, Git, CMake, PostgreSQL, Docker, AWS, MATLAB, Arduino/Embedded Systems